# EPIDEMIC OF PATHOLOGIC MYOPIA

## What Can Laboratory Studies and Epidemiology Tell Us?

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**Purpose:** To systematically review epidemiologic and laboratory studies on the etiology of high myopia and its links to pathologic myopia.

**Methods:** Regular Medline searches have been performed for the past 20 years, using "myopia" as the basic search term. The abstracts of all articles have been scrutinized for relevance, and where necessary, translations of articles in languages other than English were obtained.

**Results:** Systematic review shows that there is an epidemic of myopia and high myopia in young adults in East and Southeast Asia, with similar but smaller trends in other parts of the world. This suggests an impending epidemic of pathologic myopia. High myopia in young adults in East and Southeast Asia is now predominantly associated with environmental factors, rather than genetic background. Recent clinical trials show that the onset of myopia can be reduced by increasing the time children spend outdoors, and methods to slow the progression of myopia are now available.

**Conclusion:** High myopia is now largely associated with environmental factors that have caused the epidemic of myopia in East and Southeast Asia. An important clinical question is whether the pathologic consequences of acquired high myopia are similar to those associated with classic genetic high myopia. Increased time outdoors can be used to slow the onset of myopia, whereas methods for slowing progression are now available clinically. These approaches should enable the current epidemics of myopia and high myopia to be turned around, preventing an explosion of pathologic myopia.

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It is now generally recognized that there is an epidemic of myopia in the developed countries of East and Southeast Asia,[1] specifically Singapore,[2] Japan,[3] South Korea,[4–6] Taiwan,[7] Hong Kong,[8] and some parts of mainland China,[9–12] where 80% to 90% of young adults are now myopic. This prevalence of myopia will gradually spread through the adult population as the present and future cohorts of young adults age. In contrast, 50 years to 60 years ago, the prevalence was of the order of 10% to 30%.[5,13–18] The prevalence of myopia is also increasing in other parts of the world,[19–22] but the problem is much less extreme than in East and Southeast Asia. Nevertheless, all the evidence suggests that the population prevalence of myopia will increase in most parts of the world over the next few decades, particularly as underdeveloped countries start systematic economic development involving mass intensive education.

Less attention has been paid to the fact that the prevalence of high myopia (variably defined, but generally with a cutoff of −5 or −6 diopters) has also been increasing, although the evidence has been available for at least 10 years.[7] In the "high prevalence of myopia" locations, the prevalence of high

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myopia is now more than 10% and approaching 20% in young adults.[2,4,9,12] Clearly, this poses a major public health challenge, because of the links between high myopia and pathologic myopia, visual impairment, and blindness. Given that clinical treatment of myopic pathology is often difficult and expensive, we suggest that effective prevention of myopia and high myopia is now an important priority.

## Laboratory Approaches to Pathological Myopia

At present, there is no good animal model for studies of high myopia. The human sclera, like all mammalian sclerae, consists almost exclusively of fibrous rather than cartilaginous tissue, which contrasts with the predominantly cartilaginous sclera of lower vertebrates, including the most commonly used animal model for studies on myopia, the chicken.[23–25] This limits potential animal models to mammals. In addition, refractive development in humans follows a very distinctive pattern, in which a tight (leptokurtotic) distribution of refraction appears in the first two years after birth.[26–29] At the end of this period, the cornea reaches stability while loss of lens power and axial elongation continues, and it is only after the age of around six years that significant myopia starts to appear in most, if not all, locations.[30] This developmental pattern is not found in most mammals, although it is found in nonhuman primates.[31]

One of the features of pathologic myopia in humans is that there is a considerable time gap, covering several decades, between the appearance of high myopia and the emergence of pathologic signs. It seems likely that research on nonhuman primates will therefore need to be conducted over similar time frames, and because primate research is both costly and ethically contentious, the emergence of pathologic myopia from high myopia will probably be mainly studied in humans. The only exception might be for studies on the early stages of the development of interventions involving pharmacological agents, or scleral strengthening, where some experimental studies may be useful.

## Epidemiologic and Genetic studies

In contrast to the absence of evidence from laboratory studies, there are important lessons to be learned from epidemiologic and genetic studies. These studies have already set the scene for public health and clinical initiatives to prevent the development of high myopia, and will hopefully limit the spread of pathologic myopia.

### *The Links Between High Myopia and Pathologic Myopia*

Pathologic myopia covers a wide range of conditions, which are covered in detail in the recent book *Pathologic Myopia*.[32] Generally, high myopia seems to be an essential precursor of most forms of pathologic myopia, because there is a much greater risk of pathology once refractions are in the high myopia range than with mild or moderate myopia,[33] and the risk of pathology further increases with the severity of high myopia. The emergence of pathologies also seems to depend on age, and older myopes with higher levels of myopia are at particularly high risk of pathology.[16] Preventing high myopia in the first place, which means reducing both the onset of myopia and its progression, may thus prove to be the most effective way of dealing with the looming epidemic of high myopia in older people.

The mechanisms linking high myopia and pathologic myopia are not clear. It may be that largely mechanical factors such as the physical strain of spreading a mature retina over an enlarged sclera, with thinning of the retina and choroid, and the thinning of the sclera may predispose individuals to critical changes such as the rupture of Bruch membrane and scleral distortions such as staphyloma. It is important to note that the simple mechanisms discussed above do not preclude a role for genetic and/or environmental factors increasing the risk of developing pathology in some highly myopic individuals. There is, for example, some evidence of genetic associations with pathologic myopia involving immunologic responses,[34] and individual differences in immunologic responses could clearly play a role in translating initial pathologic lesions into full-blown sight-threatening pathology.

### *The Etiology of Myopia*

To develop preventive approaches requires an understanding of the factors that lead to the development of high myopia. Myopia was once regarded as under tight genetic control, and indeed, there are more than 200 defined familial syndromes in which myopia is an associated feature. Most of these have established chromosomal localizations, although identified genes are rare. Collectively, these forms of myopia, in which inheritance clearly plays a major role, probably account for myopia in at most 1% to 2% of the population, and while the evidence is limited, there is no evidence that the prevalence of these forms of myopia is increasing, or that the mutations identified make an important contribution to the major and increasing form of myopia, school myopia.

The etiology of school myopia has been extremely controversial.[35] For many years, ophthalmologists and optometrists were taught that myopia and ocular biometry were genetically determined, and that the only question remaining was whether environmental factors had any role.[36] However, in the last decade, this idea has been increasingly questioned. It is now clear that this formulation should be reversed, with environmental factors playing a major role in the etiology of myopia, because the rapid increases in the prevalence of myopia that have been observed are simply incompatible with genetic determination, because gene pools cannot change that fast.[1,37]

In the past 10 years, it has also become clear that modern genetic approaches have been unable to identify the number and magnitude of genetic associations expected from the consistently observed high heritability reported in twin studies. The most recent large-scale genomewide association studies have identified around 50 genetic associations, but each is of limited effect and currently account for less than 10% of phenotypic variation, although application of the new genome-wide complex trait analysis approach suggests that 25% to 35% can be accounted for by single-nucleotide polymorphisms, many of which do not show genomewide significant associations.[38–40] This gap between expectations from twin studies and results from genomewide association studies has come to be known as "missing heritability," and is a common feature of many complex diseases.[41] These limited effect sizes contrast with the 2-fold to 4-fold increases in the prevalence of myopia in East and Southeast Asia over the past few decades that must be driven by environmental change.

### *The Etiology of High Myopia*

In the case of high myopia, there has traditionally been a strong emphasis on genetic factors,[37,42,43] consistent with evidence that many of the clearly genetic forms of myopia result in high myopia. This was the case in previous generations. But the recent rapid increases in the prevalence of high myopia raise exactly the same issue posed by the rapid increases in the overall prevalence of myopia, namely, that the increase in high myopia cannot be explained in terms of genetic change, because gene pools cannot change so fast. This leads to the conclusion that we need to distinguish between two forms of high myopia—the classic form, which is substantially genetic in origin, and a newer form of high myopia, which is environmentally driven. This will almost certainly be the dominant form of high myopia in younger adults in East and Southeast Asia, and the consequences of this form of high myopia will spread throughout the population as the population ages.

There is good reason to believe that the recent increases in the prevalence of high myopia are simply a natural consequence of the increasing prevalence of ordinary myopia. As shown in Figure 1, high myopia in children and adolescents tends to remain low in prevalence (<1 to 2%) from birth until the age of 11 years to 13 years. But after this age, the prevalence of high myopia increases progressively, and probably only stabilizes in the third decade of life.

This pattern of development can be readily explained by the increasingly early age of onset of myopia,[7] as seen in Figure 1. In addition, the rate of progression of myopia decreases with age.[44] Thus, the epidemic of myopia creates a situation in which children with earlier onset myopia have longer to progress until refraction stabilizes, and they start progressing at younger ages, when progression is faster. A child who becomes myopic at the age of 6 years, with a mean annual progression rate of $-1$ D, will obviously reach a high myopia cutoff at around the age of 11 years to 13 years. This tendency to more rapid increases in the prevalence of myopia may be exacerbated if myopia progression rates are higher in populations or locations in which there is a higher prevalence of myopia, as appears to be the case.[45]

This understanding of how acquired high myopia develops means that as the prevalence of myopia

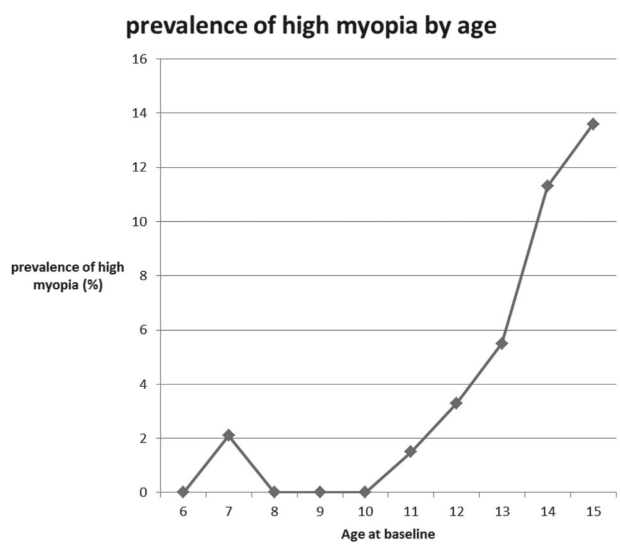


**Fig. 1.** Prevalence of high myopia at baseline from the Guangzhou Twin Eye Study. Prevalence of high myopia was defined using a cutoff of $-5$ diopters for high myopia in either eye.

increases, the prevalence of high myopia will increase disproportionately, and thus a higher proportion of the people with myopia will be highly myopic, with a correspondingly disproportionate increase in the burden of disease. This trend can be clearly seen in Figure 2.

It is likely that the predominant form of high myopia in young adults in East and Southeast Asia is now acquired as an extension of the epidemic of ordinary myopia, and is thus driven by environmental factors, in contrast to the predominantly genetic basis for high myopia in the past. The Zhongshan Ophthalmic Center—Brien Holden Vision Institute High Myopia Registry has therefore been established in Guangzhou, and has enrolled a large number of patients with high myopia aged between 7 years and 70 years. It will document how pathologic myopia develops from high myopia in this new, acquired form of high myopia.

## Environmental Factors in the Etiology of Myopia

If the current epidemic of high myopia is predominantly driven by environmental change, leading to a higher incidence of early-onset myopia and perhaps higher progression, the key question becomes, which environmental changes are involved? Many factors have changed in East and Southeast Asia since the end of the Second World War, but two factors stand out as potentially causal. A link between myopia and education is based on an extensive and consistent body of evidence, from Kepler's initial speculations,[46] to very recent studies.[37,47–49] There is strong evidence that those who have more years of education, or achieve higher grades (which are overlapping influences), are more myopic on average,

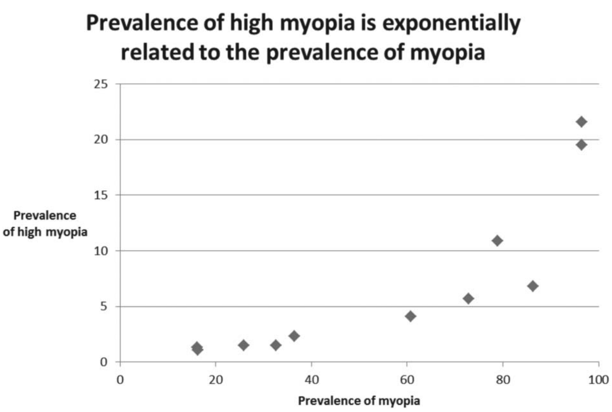


**Fig. 2.** Relationship between prevalence of myopia and prevalence of high myopia. The prevalence of high myopia increases exponentially with increasing prevalence of myopia, with rapid increases in the prevalence of high myopia occurring once the myopia prevalence reaches over 60%.

in all the ethnic groups that have been studied, making sensitivity to educational pressures an almost universal characteristic.[37] Probably only those who remain highly hyperopic throughout childhood are immune to these pressures.

How education leads to myopia is not well established. Nearwork, obviously an important component of education, has been postulated to be the link, but it remains controversial.[50] Initial thinking stressed a link between accommodative load and the development of myopia, and this appealing idea appeared to be supported by the early successful use of atropine eye drops to control myopia progression.[51] However, in animal models, atropine can block axial elongation even when it does not block accommodation, suggesting that the site of action of atropine in inhibiting axial growth lies elsewhere.[52,53]

This idea has therefore been modified to stress the importance of accommodative lag, which produces hyperopic defocus during reading. Experimental studies on young animals have shown that minus lenses (which impose hyperopic defocus) promote axial elongation, whereas plus lenses (which impose myopic defocus) slow axial elongation. Thus, it has been suggested that the hyperopic defocus associated with accommodative lag would promote axial elongation. This idea formed the basis for a major trial of progressive addition lenses to control myopia progression, which provided only small and clinically insignificant effects.[54] Accommodative lag is higher in myopes, but it is currently not clear whether this is a consequence of myopia, or whether it precedes the onset of myopia, and thus could have a causal role.[55–58] The most recent study[58] suggests that increased accommodative lag only appears after the onset of myopia.

Even the link of myopia to nearwork has been questioned.[50] Detailed longitudinal studies by the Orinda Longitudinal Study of Myopia/Collaborative Longitudinal Evaluation of Ethnicity and Refractive Error group have shown only limited effects of nearwork,[59,60] indicating that nearwork may have little effect on the development of myopia. However, a number of studies have reported more significant effects, including the Sydney Myopia Study and its five-year follow-up study the Sydney Adolescent Vascular Eye Study. There were strong associations with myopia before adjustment, and adjusted associations of myopia with habitual viewing distance and duration of reading without a break with myopia remained significant.[61] The same finding has been reported in the Anyang Childhood Eye Study.[62] But again, it is not clear whether these associations are causal.

There has also been considerable speculation about a protective effect of being outside, but it is only in the last decade that protective effects of spending time outdoors have been unequivocally demonstrated in both cross-sectional and longitudinal analyses from the Orinda Longitudinal Study of Myopia/Collaborative Longitudinal Evaluation of Ethnicity and Refractive Error[59,60] and Sydney Myopia Study/Sydney Adolescent Vascular Eye Study groups.[63–65] These findings have been supported by a large number of other papers.

The effects of spending time outdoors are powerful. Jones et al[59] showed that children who spent more than 10 hours to 14 hours outdoors a week (outside of school hours) had baseline risk of myopia, even if they had one or two parents who were myopic. Rose et al[65] showed that Chinese children growing up in Sydney reported more time outdoors than Chinese children growing up in Singapore with an almost 10-fold higher prevalence of myopia in Singapore. Within the Sydney Myopia Study/Sydney Adolescent Vascular Eye Study, the children most at risk of myopia were consistently those who combined a lot of nearwork with little time outdoors.[63,64] Thus, more time outdoors was able to neutralize the myopigenic effects of both parental myopia, and of large amounts of nearwork.

In the initial paper on the protective effect of time outdoors from the Sydney Myopia Study,[64] we proposed that bright light outdoors stimulated the release of dopamine from the retina, which inhibited axial elongation, based on experimental evidence that light stimulated dopamine release from the retina,[66–68] and that dopamine agonists reduced axial elongation.[69–71] This hypothesis has received strong experimental support, because increasing light intensity completely prevented the development of form-deprivation myopia, at light intensities well within those often encountered in human environments.[72–74]

The hypothesis that ultraviolet (UV) rather than visible light exposure caused this effect was a plausible alternative,[75–77] but it has not received much experimental support, because the bright lights used in experimental studies of protection were generally UV free,[72–74] and bright UV light does not affect emmetropization.[78] Myopic children are outside less, but lower vitamin D levels in myopes than non-myopes does not demonstrate a causal role for vitamin D.[79,80] A detailed analysis from the ALPAC study[81] provides strong evidence against the UV hypothesis. Another alternative hypothesis invoked a role for the undoubted difference in dioptric loads between indoor and outdoor environments,[33] but experimental protection under standardized laboratory conditions argues against this hypothesis.

The major argument now used to cast doubt on the light–dopamine hypothesis is that light does not protect from experimental lens-induced (defocus-induced) myopia,[74,82] which many believe to be the best model for human refractive development. However, we have argued that while the defocus model appears to fit well with human refractive development in the first two years of life, when human myopia is rare, defocus mechanisms, particularly the effects of myopic defocus, appear to be much weaker in later human refractive development, when human myopia does develop.[83] To give just one clinically relevant example of the problems with the defocus model, if the defocus model is applied to later human refractive development, given the powerful inhibitory effects of myopic defocus on axial elongation in animal studies,[84–86] myopia should be a self-limiting condition. But myopia is clearly not a self-limiting condition, and progression is the rule in corrected and uncorrected myopes. Furthermore, if myopic defocus played a major role in slowing eye growth at this stage, then correction of myopia would be harmful. However, it has been clearly demonstrated that correction of myopia is not harmful, and under-correction may possibly increase, rather than decrease progression.[87]

### *Does Time Outdoors Slow Myopia Progression?*

In practice, complete prevention of the onset of myopia may be difficult to achieve. Therefore, prevention of progression is also likely to be important for the control of high myopia. Somewhat surprisingly, because axial elongation underlies both onset and progression, the Orinda Longitudinal Study of Myopia/Collaborative Longitudinal Evaluation of Ethnicity and Refractive Error and Sydney Myopia Study/Sydney Adolescent Vascular Eye Study have not demonstrated protective effects of time outdoors on the progression of myopia.[63,88]

However, interpretations of this failure differ. One interpretation is that the mechanisms resulting in onset and progression are different. We find it difficult to understand how this could be the case, because atropine blocks both the progression and the onset of myopia in humans,[89,90] making a fundamental distinction unlikely.

We suggest that this paradoxical result may be a statistical problem. Studies on progression have to be carried out on myopic children, who spend less time outdoors and more time on nearwork, and if the range of variation in a parameter is reduced, it becomes harder to achieve statistically significant associations. In contrast, seasonal variations of progression in myopic children show higher rates of progression[91–93] in winter and lower in summer that are consistent with reported seasonal differences in time outdoors and nearwork loads,[94] and show that progression is potentially regulated by the same environmental exposures that regulate onset of myopia.

Clearly, more needs to be done to resolve this issue. But fortunately, even if it ultimately transpires that progression is not regulated by time outdoors, there are several other approaches to slowing protection. The best documented is the use of low-dose (0.01%) atropine eye drops, which appears to be largely free of short-term side effects, although there is still some uncertainty about the long-term stabilization of refraction in eyes that have been treated for prolonged periods of time with atropine.[95] The use of orthokeratology lenses also seems to slow myopic progression,[96,97] although there are some limits to studies conducted thus far and important potential safety issues. Finally, a wide range of customized spectacle and contact lenses seem to have generally modest effects on progression, which require further validation.[98]

### *Preventing Myopia With Increased Time Outdoors*

Despite the uncertainties discussed above, the evidence on the protective effects of increased time outdoors has been rapidly translated into practice. Four clinical trials have now reported results.

One was a very small trial carried out in Changsha, China over two years.[99] It limited nearwork for a group of 80 myopic children to <30 hours a week, while keeping time spent outdoors to over 14 hours to 15 hours a week. The rate of progression was reduced by 27% in the intervention group.

A major trial was carried out in Kaohsiung, Taiwan, involving 571 primary school children in 2 matched schools.[100] The intervention involved locking the children out of their classrooms at school recess times, allowing potentially an additional 80 minutes of time outdoors. This produced a 53% reduction in incident myopia, but no effects on progression.

A third trial was carried out in Sujiatun District in northern China over 1 year.[101] Over 3,000 primary school-age students were initially enrolled. Two extended recess breaks added 40 minutes outdoors to the school day, and a significantly reduced decline in uncorrected visual acuity (presumably myopia) was detected. Only 391 students received cycloplegic refractions (under 20% of the initial cohort) and this subgroup analysis suggested a 59% reduction in incident myopia. No significant effect on progression was detected.

A fourth trial was a cluster-randomized trial conducted in 12 primary schools in Guangzhou in southern China, involving around 900 students in each group.[102] The intervention was an additional 40 minutes of time outdoors, added to the end of the school day, and the intervention was tightly monitored. This produced a 23% reduction in incident myopia over 3 years. The biggest limitation of this trial was the limited nature of the intervention, which had to be approved by the Ministry of Education.

These trials have significant problems in terms of sample and intervention size. But, they are generally consistent in showing a reduction in incident myopia. The two best designed trials, those conducted in Kaohsiung and Guangzhou, gave rather consistent effects, with signs of a dose–response curve consistent with expectations from epidemiologic studies.

## Conclusions

While it is still early days in this area of research, we now have proof of principle that the onset of myopia can be reduced by increasing the amount of time that children spend outdoors. School-based interventions to increase time outdoors are preferable, because they are effectively mandatory, whereas relying on parents to change their patterns of behavior and that of their children appears to have little chance of success. It is currently not clear whether these interventions directly slow progression, but even if they do not, any delay in the onset of myopia should reduce the prevalence of high myopia. In addition, clinical interventions using the array of approaches that are now available should also help to reduce progression toward high myopia, and as a result contribute to reducing pathologic myopia and visual impairment.

**Key words:** myopia, high myopia, pathology, prevention, time outdoors, atropine.

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